



REVIEW

Biological control of the diamondback moth, *Plutella xylostella*: A review

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Abstract

The diamondback moth (DBM), *Plutella xylostella* (L.) (Lepidoptera: Plutellidae), is one of the most destructive cosmopolitan insect pests of brassicaceous crops. It was the first crop insect reported to be resistant to DDT and now, in many crucifer producing regions, it has shown significant resistance to almost every synthetic insecticide applied in the field. In certain parts of the world, economical production of crucifers has become almost impossible due to insecticidal control failures. Consequently, increased efforts worldwide have been undertaken to develop integrated pest management (IPM) programs, principally based on manipulation of its natural enemies. Although over 130 parasitoid species are known to attack various life stages of DBM, most control worldwide is achieved by relatively few hymenopteran species belonging to the ichneumonid genera *Diadegma* and *Diadromus*, the braconid genera *Microplitis* and *Cotesia*, and the eulophid genus *Oomyzus*. DBM populations native to different regions have genetic and biological differences, and specific parasitoid strains may be associated with the specific DBM strains. Therefore, accurate identification based on genetic studies of both host and parasitoid is of crucial importance to attaining successful control of DBM through inoculative or inundative releases. Although parasitoids of DBM larvae and pupae are currently its principal regulators, bacteria-derived products (e.g., crystal toxins from *Bacillus thuringiensis*) and myco-insecticides principally based on *Zoophthora radicans* and *Beauveria bassiana* are increasingly being applied or investigated for biological control. Viruses, nematodes and microsporidia also have potential as biopesticides for DBM. When an insect pest is exposed to more than one mortality factor, there is the possibility of interactions that can enhance, limit, or limit and enhance the various aspects of effectiveness of a particular control tactic. This paper reviews the effectiveness of various parasitoids and entomopathogens against DBM, interactions among them, and their possible integration into modern IPM programs.

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Introduction

Brassicaceae (=Cruciferae) comprises a diverse group of 350 genera and over 3,500 species of dicotyledonous cultivated and wild herbs characterized by hypogynous, complete and cruciform flowers (Warwick et al. 2003). Members of this widespread family represent economically important crops like oilseeds (e.g., canola and mustard), cole crops (e.g., cabbage and cauliflower) and root vegetables (e.g., radish and turnip). They occur in tropical and temperate climates and are the most common and important components in the diets of various cultures (Shelton 2004). In 2004, about 3.24 million ha of cabbage crop and 893,216 ha of cauliflower crop were harvested worldwide (FAO 2004). Oilseed *Brassica* species supply 14% of global vegetable oil production and rank third in importance after soybean and palm (Murphy 1996).

The diamondback moth (DBM), *Plutella xylostella* (L.) (Lepidoptera: Plutellidae), is one of the most destructive cosmopolitan pests of cruciferous crops that requires US\$1.0 billion globally in estimated annual management costs (Talekar & Shelton 1993) in addition to the crop losses. This crucifer specialist may have its origin in Europe (Hardy 1938), but on the basis of the presence of its biocontrol agents (14 species of parasitoids) and host plants (175 species, of which 32 are exotic), Kfir (1998) speculated that it originated in South Africa. Using similar arguments, Liu et al. (2000) are of the view that *P. xylostella* originated in China. This pest is now present wherever its host plants exist and is considered to be the most widely distributed of all Lepidoptera (Shelton 2004).

Outbreaks of DBM in Southeast Asia sometimes cause crop losses of more than 90% (Verkerk & Wright 1996). It is a very destructive pest in southern Sind (Hyderabad and Karachi region) of Pakistan where cruciferous vegetables are grown throughout the year and due to its severe infestations, growers are sometimes compelled to plough down their standing crops in spite of multiple insecticide applications (Abro et al. 1994). It occurs throughout the United States (Harcourt 1957) and is one of the major pests of crucifers in various regions including the southeast and Pacific Northwest (Buntin 1990; Brown et al. 1999). In southeastern USA, it constitutes >90% of the guild of defoliating lepidopterans of canola and can occur from seedling through crop maturity (Buntin 1990; Ramachandran et al. 2000). DBM is also a serious pest of crucifers in Canada. It was likely introduced from Europe before 1854 and now occurs annually throughout the Canadian prairies wherever brassicaceous plants are grown (Anonymous 1996; Dosdall et al. 2004b). In 1985, about 467,860 ha were treated with insecticides in Alberta and Saskatchewan to control DBM with an estimated cost of CN\$11.9 million (Madder & Stemeroff 1988). In 2001, about 1.8 million ha were sprayed with insecticides in western Canada to control various insect pests primarily the DBM (WCCP 2001). The source of DBM in western Canada is attributed to influxes from southerly regions (Smith & Sears 1982; Dosdall et al. 2004b). Similar migrations have been reported in the UK (Hardy 1938; Chapman et al. 2002), New Zealand, South Africa, and southern parts of Chile and Argentina (Talekar & Shelton 1993), Japan (Honda 1992) and Australia (Goodwin & Danthanarayana 1984). Moths are highly migratory and have been recorded to travel a distance of about 1500 km at 400–500 km per night (Chapman et al. 2002).

The DBM is multivoltine with four to 20 generations per year in temperate and tropical regions, respectively (Harcourt 1986; Vickers et al. 2004). Each female can lay over 200 eggs mainly on the upper leaf surface (Talekar et al. 1994; Justus et al. 2000). Of 896 eggs oviposited on Chinese cabbage, 68% were laid on upper leaf surfaces, 37% on lower leaf surfaces and 3% on stems (Talekar et al. 1994). Eggs hatch after 4–8 days and first instar larvae usually mine the spongy mesophyll tissues (Harcourt 1957). Second, third and fourth instar larvae are surface feeders and they consume leaves, buds, flowers, siliques, the green outer layer of stems and also developing seeds within older siliques (Anonymous 1996). The average duration of larval instars under Canadian field conditions was 4.0, 3.6, 3.4, and 4.2 days for the first through fourth instars, respectively and pupation required 7.8–9.8 days (Harcourt 1957). The observation that as many as 20 generations per year can occur in the tropics indicates that developmental time from egg to adult is much more rapid in those regions. Successful development can occur at constant temperatures from 8 to 32°C and under alternating regimes including temperature as low as 4°C or as high as 38°C (Liu et al. 2002). There is also evidence for its overwintering in certain temperate parts of the world including the USA (Marsh 1917), the UK (Hardy 1938) and Canada (Dosdall 1994).

Host plant availability and action of its natural enemies are two key biotic factors that regulate DBM populations in the field (Harcourt 1986; Kfir 1997). In many countries, synthetic insecticides are used to control DBM, which often eliminates natural enemies. This, in turn, can lead to continued intensive use of insecticides, eventual insecticide resistance and control failure. DBM was the first crop insect to be reported resistant to DDT in 1953 in Java, Indonesia (Ankersmit 1953), and now in many crucifer-producing regions it has shown significant resistance to almost every insecticide applied in the field including new chemistries such as spinosyns, avermectins, neonicotinoids, pyrazoles and oxadiazines (Sarfranz & Keddie 2005; Sarfranz et al. 2005). This has prompted increased efforts worldwide to develop integrated pest management (IPM) programs, principally based on manipulation of natural enemies.

DBM natural enemies: Predators and parasitoids

In an insightful statement, Marsh (1917) outlined the pest status of DBM in the USA and declared, 'The diamondback moth is a striking example of a potentially serious pest normally held in repression by parasites'. On a worldwide scale, over 135 parasitoid species have been recorded to attack various stages of DBM (Delvare 2004), with the most common ones comprising six species of egg parasitoids, 38 larval, and 13 pupal parasitoids (Lim 1986; Talekar & Shelton 1993) (Figure 1). Certain ants, flies, lacewings, hemipterans, beetles, spiders and birds also prey on its larvae (Alam 1992; Anonymous 1996; Reddy et al. 2004; Endersby & Cameron 2004). Some predators such as *Chrysoperla carnea* Stephens (Neuroptera: Chrysopidae) were attracted to DBM pheromone blend, larval frass and green leaf volatiles of crucifers (Reddy et al. 2004). These generalist predators have not usually been considered significant factors in regulation of DBM populations although some ants such as *Anomma nigricans* (Illiger) (Hymenoptera: Formicidae) are major control agents in periurban areas of Benin, West Africa (Goudegnon et al. 2004). Egg parasitoids (genera *Trichogramma* and *Trichogrammatoidea*) (Hymenoptera:

Trichogrammatidae) do not exert adequate control, as they require frequent mass releases (Talekar & Shelton 1993). In addition, egg parasitoids are not always host-specific (Goulet & Huber 1993) and may pose a threat to non-target species in a region. For example, *Trichogramma brassicae* Bezdenko, which was inundatively released against the European corn borer in Switzerland, parasitized eggs of 22 out of 23 lepidopteran species tested, including those on the Swiss list of endangered species (Babendreier et al. 2003a,b). Larval parasitoids have the greatest control potential and Lim (1986) suggested that the most effective belong to the hymenopterian genera *Microplitis* (Braconidae), *Cotesia* (Braconidae), and *Diadegma* (Ichneumonidae). A few prepupal and pupal parasitoids of the genus *Diadromus* (Ichneumonidae) also contribute to DBM control (Table I) (Hardy 1938; Liu et al. 2000; Kirk et al. 2004), and occasionally *Pteromalus* (Hymenoptera: Pteromalidae) species also parasitize DBM pupae (Chauhan & Sharma 2004).

Mustata (1992) reported 25 parasitoid species from Moldavia (Romania) parasitising up to 80–90% of DBM populations but some of them were misidentified (see Delvare 2004) and others were rejected by Noyes (1994) as they were based on inaccurate data. Kirk et al. (2004) recorded 27 primary parasitoid species mainly of genera *Diadegma*, *Cotesia* and *Oomyzus* on 115 populations of DBM collected in 32 countries. In Ethiopia, *Diadegma* spp. and *Cotesia plutellae* (Kurdj.) were responsible for over 90% of observed parasitism (Ayalew et al. 2004). *Cotesia plutellae* and *Oomyzus sokolowskii* (Kurdj.) (= *Tetrastichus sokolowskii*) (Hymenoptera: Eulophidae) are considered the most predominant parasitoids in China (Liu et al. 2000). *Diadegma* spp. and *Diadromus* spp. predominate in Europe (Hardy 1938), South Africa (Kfir 1997) and New Zealand (Todd 1959). *Diadegma insulare* (Cress.) caused about 71% parasitism in California (Oatman & Planter 1979). *Diadegma semiclausum* (Hellén), *D. rapi* (Cameron) and *Diadromus collaris* (Grav.) collectively accounted for 93% parasitism in Victoria, Australia (Goodwin 1979). Seven species were observed in Brasilia with *D. leonitinae* (Brethés) and *Apanteles piceotrichosus* (Blanchard) as dominant parasitoids but the total parasitism was insufficient to regulate DBM populations (Guilloux et al. 2004). In the Eastern Cape (South Africa), four species viz., *Diadegma mollipla* (Holmgren), *C. plutellae*, *D. collaris* and *O. sokolowskii* were major parasitoids and DBM parasitism rates varied throughout the year ranging from 10 to 80% (Smith & Villet 2004).

Canada has three major hymenopterous parasitoids of DBM. *Diadegma insulare* (Cress.) and *Microplitis plutellae* (Mues.) are larval parasitoids while *Diadromus subtilicornis* (Grav.) attacks the prepupal and pupal stages (Harcourt 1986; Anonymous 1996; Braun et al. 2004; Dossdall et al. 2004b). In Alberta, *D. insulare* was the principal parasitoid of DBM in 1992, accounting for approximately 45% of the total parasitism, while *M. plutellae* and *D. subtilicornis* each accounted for about 14% of the total parasitism. A similar situation was observed in Saskatchewan with 30% parasitism by *D. insulare* and <8% each by *M. plutellae* and *D. subtilicornis* (Braun et al. 2004). Shelton et al. (2002) collected 2,815 larvae and pupae in New York from 1979 to 1994 and observed that 46.5% of them were parasitized by *D. insulare*, 7.0% by *M. plutellae*, 2.1% by *O. sokolowskii*, and 0.4% by *D. subtilicornis*. In contrast, the absence of effective parasitoids in areas including Southeast Asia, the Pacific islands, the Caribbean, and Central America can result in intensive DBM infestations (Talekar & Shelton 1993).

Table I. Biodiversity and developmental parameters associated with major parasitoids of *P. xylostella*.

Parameters	<i>Diadegma</i> species	<i>Microplitis</i> species	<i>Cotesia</i> species	<i>Oomyzus</i> species	<i>Diadromus</i> species	References
Number of reported species commonly associated with DBM	9	2	1	1	2	Hardy (1938); Talekar and Shelton (1993); Azidah et al. (2000); Kirk et al. (2004) and others
Host stage attacked	Larval	Larval	Larval	Larval–pupal	Prepupal–pupal	Lloyd (1940); Xu et al. (2001); Shi et al. (2004)
Life style	Solitary	Solitary	Solitary	Gregarious	Solitary	Lloyd (1940); Putnam (1968); Ooi (1988); Rowell (2004)
Adult body size (mm)	5–7	2.3–2.5	2.5–2.6	1.3–1.4	6–7	Gharuka et al. (2004); Rowell (2004)
Evidence for pesticide resistance	Yes	?	Yes	?	?	Xu et al. (2001); Liu et al. (2004); Rowell (2004)
Number of larval instars	4	3	4(?)	?	?	Bolter and Laing (1983); Shi et al. (2002); Gharuka et al. (2004)
Average developmental time (egg to adult) (days)	12 to 18	12 to 18	13 to 18	13 to 17	15 to 17	Bolter and Laing (1983); Ferreira and Torres (2003); Talekar (2004); Rowell (2004)
Optimum developmental temperature (°C)	20 to 22	20 to 22	26 to 28	26 to 28	22 to 26	Bolter and Laing (1983); Liu et al. (2001); Talekar (2004)
Average thermal requirement	282 °D _{6.6} (<i>D. insulare</i>)	218 °D _{9.2} (<i>M. plutellae</i>)	?	211.8 °D _{11.6} (<i>O. sokolowskii</i>)	225.1 °D _{7.4} (<i>D. collaris</i>)	Bolter and Laing (1983); Liu et al. (2001); Ferreira and Torres (2003)

Larval parasitoids

Diadegma insulare (Cresson) (Hymenoptera: Ichneumonidae). *Diadegma* species are the principal regulators of *P. xylostella* worldwide (Table II) (Marsh 1917; Hardy 1938; Todd 1959; Mustata 1992; Azidah et al. 2000; Delvare 2004). This large genus of Campopleginae comprises about 200 species worldwide (Yu & Horstmann 1997), with the majority (about 118 species) being described from temperate regions (Delvare 2004).

Diadegma insulare is a solitary, host-specific larval endoparasitoid of DBM and is one of its most important biocontrol agents in the Nearctic to the northern Neotropical regions (Harcourt 1960, 1986; Fitton & Walker 1992; Mukenfuss et al. 1992; Idris & Grafius 2001). On average, it can parasitize 70–90% of DBM larvae and parasitized larvae consume 35–80% less food than nonparasitized larvae (Mukenfuss et al. 1992; Mitchell et al. 1997; Sourakov & Mitchell 2000; Monnerat et al. 2002). In North America, parasitism by *D. insulare* sometimes exceeds 80% for fourth instars and 50% for third instars (Hutchison et al. 2004). In south Texas, it has been reported to account for >98% parasitism in the field (Legaspi et al. 2000). Compared with other parasitoids, it is an efficient host searcher (Xu et al. 2001; Wang & Keller 2002) and has a significant ability to discriminate previously parasitized hosts (i.e., it avoids multiparasitism and superparasitism) (Bolter & Laing 1983). During a search immediately following contact by a parasitoid, the DBM larva typically moves away wriggling vigorously and drops from the plant suspended by a silken thread. A wasp may attempt a quick jab with its ovipositor and sometimes inserts an egg successfully.

Table II. *Diadegma* species known to parasitize the DBM throughout the world.

<i>Diadegma</i> species	Distribution	References
<i>D. insulare</i> (Cresson) (<i>D. plutellae</i> , <i>D. hellulae</i> , <i>D. congregator</i> , <i>D. polynesialis</i> , <i>D. pygmaeus</i>)	Nearctic to the northern Neotropical region and some Pacific islands, including Hawaii (only one species is known to attack DBM in North America)	Fitton and Walker (1992); Azidah et al. (2000); Kirk et al. (2004); Wagener et al. (2004b)
<i>D. fenestrale</i> (Holmgren) (= <i>D.</i> <i>varuna</i> , <i>D. niponica</i> , <i>D. gracilis</i>)	Palaearctic and south-east to Sri Lanka and the Philippines	Fitton and Walker (1992); Azidah et al. (2000); Wagener et al. (2004b)
<i>D. leontinae</i> (Br��thes)	Southern Neotropical region: Argentina, Uruguay	Fitton and Walker (1992); Azidah et al. (2000); Kirk et al. (2004); Wagener et al. (2004b)
<i>D. mollipla</i> (Holmgren)	Afrotropical, including some Indian Ocean and South Atlantic islands	Azidah et al. (2000); Kirk et al. (2004); Wagener et al. (2004b)
<i>D. exareolator</i> Aubert <i>Diadegma</i> sp. n.	Great Britain Ethiopia	Shaw and Hortsman (1997) Wagener et al. (2004b); Kirk et al. (2004)
<i>D. novaezealandiae</i> Azidah, Fitton & Quicke	New Zealand	Azidah et al. (2000)
<i>D. rapi</i> (Cameron)	South-eastern Australia	Fitton and Walker (1992); Azidah et al. (2000); Wagener et al. (2004b)
<i>D. semiclausum</i> (Hell��n) (= <i>D. xylostellae</i> , <i>D. eucerophaga</i> , <i>Angitia cerophaga</i>)	Palaearctic, and many other parts of the world	Fitton and Walker (1992); Azidah et al. (2000); Kirk et al. (2004); Wagener et al. (2004b)

Interestingly, *D. insulare* exhibits very flexible behavior sitting motionless near the silken thread, waiting for the suspended larva to climb up and then attacks it again. Sometimes the wasp also follows down the silken thread and stings the suspended larva (Sarfraz, personal observation). A similar behavior was reported for its congeneric species, *D. semiclausum* (Wang & Keller 2002).

Attempts have been made to rear *D. insulare* in captivity on DBM larvae reared on artificial diet (Sieglaff et al. 1998; Johanowicz & Mitchell 2000b) but to date no significant success has been achieved. Mass rearing presents many challenges. In the absence of a host plant (or leaf), female wasps rarely parasitize DBM larvae (Sourakov & Mitchell 2000). Under laboratory conditions, the ratio of males to females is usually extremely high (about 10:1) due to frequent haploid parthenogenesis; in the field this ratio is equivalent (about 1.1:1.0) (Harcourt 1960). The proportion of male wasps from young parasitized DBM larvae (L_2) is greater than females while larvae parasitized at L_3 and L_4 stages yield more females (Fox et al. 1990; Monnerat et al. 2002). Temperature also affects parasitism rates and sex ratio. Monnerat et al. (2002) demonstrated that *Diadegma* sp. did not parasitize at 15°C, while at temperatures of 20, 25 and 29°C, parasitism rates were similar. Sex ratios were identical at 20 and 25°C but at 29°C the ratio was 3:1 (m:f).

Diadegma insulare has been reared successfully in the greenhouse with a parasitism rate of 95% with 45–63% females (Xu & Shelton 2001). Sourakov and Mitchell (2002) developed a technique to achieve a higher female to male sex ratio and increased progeny. They offered 100+ third instar DBM larvae to a single mated female wasp in the presence of collard leaves. Every 24 h, this female parasitoid was transferred to another container with a new group of larvae for four consecutive days. The female progeny increased from $22.6 \pm 5.8\%$ (day 1) to $59.1 \pm 5.06\%$ (day 4). They suggested that oviposition should be allowed for at least 4 days in order to get a higher female:male ratio and increased total numbers of progeny.

In the field, this highly mobile wasp prefers to visit or remain longer in habitats with abundant food sources and refugia for itself and its host (Idris & Grafius 2001). Depending upon the food source, the number of DBM larvae parasitized by a single wasp may vary from zero to 150 (Grafius 1997). Many flowering plants provide good nectar sources for parasitoids in the field. Sweet alyssum, *Lobularia maritima* (L.) Desv. (Brassicaceae), for example, can be a good food source for augmentatively released parasitoids in the cabbage agroecosystem (Johanowicz & Mitchell 2000a).

Laboratory studies indicated that *D. insulare* can increase its tolerance to certain pyrethroids such as permethrin (Xu et al. 2001) but the development of resistance was much slower than its host, *P. xylostella*, and its ability to build up tolerance was limited (Shelton 2004). For instance, a *P. xylostella* population collected from California exhibited over 200-fold tolerance to permethrin (Shelton et al. 2000) while *D. insulare* developed only a 5-fold resistance to this pyrethroid in the laboratory (Xu et al. 2001).

Microplitis plutellae (Muesebeck) (Hymenoptera: Braconidae). *Microplitis plutellae* is also a primary larval endoparasitoid of DBM especially in North America. Initially it was listed only from Iowa, Colorado, Idaho, and California, but now it has also been recorded from Utah, South Carolina, New York, Alberta, Saskatchewan, Ontario (Harcourt 1960; Braun et al. 2004), Taiwan, Laos and Cambodia (Kirk et al. 2004). A congeneric species, *M. mediator* (Haliday) is also known to parasitize DBM and can be quite common in Romania (Kirk et al. 2004). *Microplitis plutellae* can overwinter in

western Canada and plays an appreciable role in regulating DBM populations early in the year (Putnam 1978). The mature parasitic larva emerges from the membranous area between the fourth and fifth abdominal tergites of the final instar host and spins its own brown, oval cocoon that adheres to any available surface (e.g., stem, leaf, wall of a cage, etc.) a short distance from the dying host (Harcourt 1960; Putnam 1968; Gharuka et al. 2004).

Comparison between D. insulare and M. plutellae. Although both *D. insulare* and *M. plutellae* parasitize all four instars of *P. xylostella* (with very low parasitism of first instars), *M. plutellae* kills and emerges from fourth instars, whereas *D. insulare* kills and emerges from the prepupal stage and spins its own cocoon inside the loosely woven cocoon of its host (Harcourt 1960; Putnam 1968). The number of generations per year of both parasitoids corresponds to the number of generations of DBM as one host larva supports only one parasitoid larva (Sourakov & Mitchell 2000). *Microplitis plutellae* can enter pupal diapause and successfully withstand low temperatures (0°C for 160 days) (Putnam 1978). In contrast, cocoons of *D. insulare* completely lose their viability within 49 days at 4°C (Okine et al. 1996; Sourakov & Mitchell 2000). On average, adult *M. plutellae* live for 20 days and produce 316 eggs per female, whereas adult *D. insulare* live for 26 days and lay 814 eggs per female at 23°C. The number of eggs produced per day is similar for both parasitoids, but adult *M. plutellae* have a shorter lifetime, resulting in higher overall fecundity for *D. insulare* (Bolter & Laing 1983).

Xu et al. (2001) collected a total of 2,512 *P. xylostella* larvae during the cabbage-growing season (from 7 June to 18 October) from the Fruit and Vegetable Crops Research Farm in Geneva (New York), with 1,040 parasitized. For the entire season, they found that the average parasitism collectively by both parasitoids (*D. insulare* and *M. plutellae*) was 33.6 and 53.6% for the third and fourth instars, respectively. For third instars, 64.9% of the total parasitism was caused by *D. insulare* and 33% by *M. plutellae*. For fourth instars, *D. insulare* accounted for 56.5% of total parasitism, whereas 41.7% was due to *M. plutellae*. During a naive wasp test, *D. insulare* was much more active than *M. plutellae*. Of tested naive wasps, 62.5% of *D. insulare* landed on the host plants within the 20-min observation period in contrast to 21.2% of *M. plutellae*. For experienced wasps, the difference between species was also highly significant, with 72.5% of experienced *D. insulare* and 40.0% of *M. plutellae* landing on host plants. Based on these results, Xu et al. (2001) suggested that *D. insulare* may be more suitable for field release to augment biocontrol against DBM.

Cotesia (=Apanteles) *plutellae* (Kurdjumov) (Hymenoptera: Braconidae). *Cotesia plutellae* is an oligophagous solitary larval endoparasitoid that parasitizes a small percentage of the larval population of DBM (Lloyd 1940). This small braconid is distributed throughout Europe (Lloyd 1940) and has also been reported from other parts of the world including China (Liu et al. 2000), South Africa (Kfir 1997), Japan (Noda et al. 1996), Pakistan (Mushtaq & Mohyuddin 1987), India (Joshi & Sharma 1974), and Indonesia (Wilkinson 1931). It has been introduced from Europe to several countries, including Australia, Dominica, Fiji, Thailand and the United States (Waterhouse & Norris 1987) and from South Africa to St. Helena (Kfir & Thomas 2001). In Pakistan, it is distributed from an altitude of 5 m (Karachi) to 1,660 m (Quetta) with highest parasitism (57.2%) of DBM recorded in turnip fields in

Hyderabad (30 m altitude) (Mushtaq & Mohyuddin 1987). In Hangzhou (China), it is a major parasitoid of DBM, remaining active throughout the year, but with very low populations in winters (Liu et al. 2000).

Cotesia plutellae shows a preference for unparasitized host larvae but has poor interspecific discrimination. When no unparasitized hosts are available, it readily oviposits in hosts containing larvae of other parasitoids such as *Diadegma* species (i.e., multiparasitism) (Lloyd 1940). Velasco (1982) reported that *C. plutellae* did not parasitize fourth instar larvae but later reports indicated that it parasitized larvae of all instars (preferably L₂ and L₃). The fourth instar larvae parasitized by *C. plutellae* exhibited longer development time (123.8 ± 1.8 h) than unparasitized ones (43.2 ± 1.1 h). In contrast to *D. insulare*, larvae parasitized by *C. plutellae* consumed more food (118.4 ± 2.4 mg) than the controls (96.2 ± 4.8 mg) (Shi et al. 2002).

Plutella xylostella and *C. plutellae* share a major metabolic mechanism of resistance (i.e., elevated levels of detoxifying enzyme, the monooxygenase) to certain pyrethroids (Liu et al. 2004) and selection of parasitoid larvae inside the resistant host can accelerate development of resistance to these insecticides in larvae and adults of *C. plutellae*. For example, the parasitoids selected with the fenvalerate-susceptible hosts acquired 4.5-fold resistance after 14 selections, while those selected with the highly resistant hosts exhibited 13.6-fold resistance to this pyrethroid after 13 selections (Liu et al. 2004). In South Africa, *C. plutellae* was abundant in the fields treated mainly with pyrethroids suggesting that it may have also developed resistance to these chemicals in the open fields (Smith & Villet 2004).

Unlike specialist parasitoids, *C. plutellae* exhibited a fixed host searching behavior, pursued the host(s) down the silk thread and spent a significantly greater proportion of time on the ground ($17.7 \pm 3.6\%$) than *D. semiclausum* ($4.7 \pm 1.1\%$) (Wang & Keller 2002). Initially, this wasp was assumed to be a DBM specialist but it has been recorded from or reared on several other lepidopterans (Fitton & Walker 1992) including a naturally occurring hybrid of *Nyctemera amica* (White) and *N. annulata* (Boisduval) (Lepidoptera: Arctiidae) (Cameron & Walker 1997). A glasshouse host specificity test in Australia revealed that *C. plutellae* had significant preference for *N. amica* on a noxious weed ragwort, *Senecio jacobaea* L. (Compositae) in the presence of *P. xylostella* larvae on cabbage (Endersby & Cameron 2004). This research suggests plant volatiles from non-crucifers (e.g., *S. jacobaea*) may be more attractive to the parasitoid than those from cabbage and introductions of *C. plutellae* to such regions may pose a threat to indigenous beneficial fauna.

Oomyzus sokolowskii (Kurdjumov) (Hymenoptera: Eulophidae). The chalcid, *O. sokolowskii* is a larval–pupal endoparasitoid commonly found in China (Liu et al. 2000, 2001; Mahmood et al. 2003; Shi et al. 2004), France, Romania, Senegal, Benin, Réunion, India, Pakistan, Brazil, Guadeloupe, Martinique, Turkey, South Africa and Italy (Kirk et al. 2004; Smith & Villet 2004). It is a gregarious parasitoid and one DBM larva can support the complete development of about eight to 10 *O. sokolowskii* (Ooi 1988; Wang et al. 1999). It has a broad host range and occasionally acts as a facultative hyperparasitoid as it sometimes emerged from cocoons of *C. plutellae* (Waterhouse & Norris 1987; Fitton & Walker 1992; Liu et al. 2000; Mahmood et al. 2003; Kirk et al. 2004). It can parasitize hosts already containing larvae of *D. semiclausum* without any discrimination but *D. semiclausum* kills the *O. sokolowskii* immatures inside the host (Shi et al. 2004).

Pupal parasitoid

Diadromus (= *Thyraella*) *collaris* (Gravenhorst) (Hymenoptera: Ichneumonidae). *Diadromus* species are primary prepupal and pupal parasitoids of DBM in various regions of the world including England (Hardy 1938), Holland (Lloyd 1940), Canada (Harcourt 1960; Anonymous 1996; Braun et al. 2004), Australia (Goodwin 1979), Moldavia (Mustata 1992), South Africa (Kfir 1997, 1998; Kirk et al. 2004), China (Liu et al. 2000), India (Chauhan & Sharma 2004), France, Turkey, Bulgaria, Georgia and Greece (Kirk et al. 2004).

Diadromus collaris is a solitary endoparasitoid that supplements the control achieved by other parasitoids. It spends its egg, larval and pupal stages inside the DBM pupa while the adult is free-living. It has been successfully introduced and established in different regions including Australia (Wilson 1960; Goodwin 1979), Barbados, New Zealand (Beck and Cameron 1990), Malaysia (Ooi & Lim 1989) and St. Helena (Kfir & Thomas 2001).

When there is opportunity, the *D. collaris* female prefers pupae that are in the first half of their pupal development, but oviposition is not restricted when no other choice is available; however, its survival decreases sharply with an increase in host pupal age (Lloyd 1940; Wang & Liu 2002). The presence of a silken cocoon surrounding the host pupa plays an appreciable role in the acceptance and parasitization of a particular host. This species exhibits superparasitism but not multiparasitism. Interspecific discrimination is pronounced and hosts containing fresh parasite eggs (24 h old) are avoided by the females irrespective of the females which laid eggs. Lloyd (1940) offered DBM prepupae containing advanced larvae of a parasitoid, *D. semiclausum* (= *Angitia cerophaga*), to ten *D. collaris* females alternately with unparasitized prepupae. A total of 75 eggs were laid; all were placed in unparasitized hosts. Similarly, Liu et al. (2001) offered DBM pupae that were parasitized as fourth instar larvae by *O. sokolowskii*. The female *D. collaris* examined these host pupae by inserting its ovipositor. However, dissection of such pupae revealed that *D. collaris* females did not deposit any eggs in the host pupae already containing *O. sokolowskii* immature stages. These secondary attacks by *D. collaris* did not show any adverse effect on *O. sokolowskii* survival (Liu et al. 2001).

Releases of parasitoids: Successes and failures

If naturally occurring biocontrol agents fail to colonize infested fields, or colonize too late in the season to provide effective pest control, augmentive/inundative releases of natural enemies may be effective for reducing pest damage (Collier & Steenwyk 2004). In contrast, inoculative releases entail the release of relatively low numbers of biocontrol agents early in the season allowing them to establish successfully for pest control later on. This strategy may prove to be an effective tool where DBM is known to overwinter and causes predictable infestations and damage later in the season.

One of the earliest parasitoid introductions was made in New Zealand in 1935 when no significant control of DBM by the indigenous parasitoids was evident. The climatic similarities of England and New Zealand prompted entomologists to introduce *D. semiclausum* and *D. collaris*, which are effective biocontrol agents in England, into New Zealand. The introduced parasitoids currently continue to suppress the pest populations along with the native *D. novaezealandiae*, but still need to be incorporated properly into modern IPM programs (Hardy 1938; Talekar & Shelton 1993). In the

early 1950s, *D. semiclausum* was introduced from New Zealand into the highlands of Java (Indonesia) (Vos 1953), where it gave significant control of DBM in conjunction with *Bt*-insecticides (Talekar & Shelton 1993). This parasitoid was subsequently also introduced from Java to the other islands in Indonesia (Talekar & Shelton 1993). When *C. plutellae* could not give adequate control, *D. semiclausum* was introduced from Indonesia to Taiwan in 1985. In the lowlands, this parasitoid failed to establish due to prevailing high temperature conditions; however, in the highlands it accounted for more than 70% parasitism (AVRDC 1986, 1988). *Microplitis plutellae* was introduced from the USA to Taiwan in 1995 (AVRDC 1998). In 2001, the International Centre of Insect Physiology and Ecology (ICIPE) imported *D. semiclausum* from Taiwan and released it in Kenya for biocontrol of DBM (Wagener et al. 2004b).

Three parasitoids viz., *D. semiclausum*, *C. plutellae* and *D. collaris* were introduced to Australia and these introductions resulted in 72–94% parasitism of DBM. Among the introduced parasitoids, *D. semiclausum* became established throughout Australia, including Tasmania, *C. plutellae* established in Australian Capital Territory, New South Wales and Queensland, and *D. collaris* established mainly in Queensland, New South Wales, Victoria, and Tasmania (Wilson 1960; Goodwin 1979).

In 1970, *C. plutellae* obtained from India was introduced to several Caribbean countries including Grenada, St. Vincent, St. Lucia, Dominica, Antigua, Montserrat, Belize, St. Kitts-Nevis, Trinidad, Barbados and Jamaica. In some release sites, the parasitoid was recovered, but it did not prove very effective in DBM suppression. Its reintroduction to Jamaica in early 1989 resulted in its establishment; parasitism increased from 5.4% in the first generation following introduction to 88.7% by March 1990 (Alam 1992; Talekar & Shelton 1993).

Between 1977 and 1978, Malaysian entomologists introduced *D. semiclausum*, *O. sokolowskii* and *D. collaris* to the Cameron Highlands where crucifers are grown year round. Although all parasitoid species were recovered the following year(s), the combined parasitism level by both *D. semiclausum* and *D. collaris* was only 6%. *Oomyzus sokolowskii* was recovered in 1978 but was not found in 1984. Surveys in 1989 revealed that *D. semiclausum* and *D. collaris* had become abundant enough to adequately suppress DBM populations (Chua & Ooi 1986; Ooi & Lim 1989; Syed et al. 1990).

Plutella xylostella is a serious pest on the island of St. Helena, a small British island in the South Atlantic Ocean. *Diadegma mollipla* (Holmgren) was the only parasitoid on the island but because of extensive use of synthetic pesticides, its impact was minimal. The Plant Protection Research Institute (PPRI) in Pretoria, South Africa introduced *C. plutellae* and *D. collaris* to St. Helena. A total of 17,500 *C. plutellae* and 23,500 *D. collaris* were released in 10 different farms across the island continuously from May 1999 to September 2000 and farmers were advised to replace chemical insecticides with *B. thuringiensis* to maximize opportunities for parasitoid establishment. During January to March 2000, the DBM larvae were sampled from 16 farms and *C. plutellae* exhibited a high parasitism level (27.7–80%) and was present in samples from 15 farms, eight of which were not released sites. *Diadromus collaris* was also recovered from five of 14 farms with a parasitism rate as high as 55%. In 2001, the use of chemical and *Bt* insecticides declined dramatically because DBM was suppressed by native and introduced biocontrol agents (Kfir & Thomas 2001).

Population genetics: Implications for biological control of P. xylostella

The success of biological control programs depends on accurate identification of the natural enemies and the linkage of host strain with coevolved natural enemies. Misidentifications or failure to match host and parasitoid genotype may lead to program failure and/or threat to native beneficial fauna. According to Noyes (1994), about 75% of host–parasitoid records are misleading because they are based on misidentifications either of hosts or their parasitoids. The main taxonomic dilemma concerns the genus *Diadegma* (Fitton & Walker 1992). For instance, in 1953, *D. insulare* from Kenya was imported and released in Hawaii, but failed to establish (Johnson et al. 1988). In 1998, the *Diadegma* species from Kenya were identified as *D. semiclausum* by the Natural History Museum, London (Wagener et al. 2004b). In 2000, all African *Diadegma* species were classified as *D. mollipla* on the basis of common morphological characters (Azidah et al. 2000) (Table II). Subsequently, Wagener et al. (2004b) separated seven different *Diadegma* species using PCR–RFLP (polymerase chain reaction–restriction fragment length polymorphism) analysis and also discovered a new *Diadegma* species from Ethiopia. DNA from the seven *Diadegma* species was successfully amplified with the internal transcribed spacers (ITS2-1 and ITS2-2), resulting in three differently sized PCR-products. In the new Ethiopian species of *Diadegma*, a DNA fragment of approximately 780 bp was obtained, while in *D. insulare*, *D. mollipla*, *D. fenestrale* and *D. semiclausum* a slightly longer band of about 800 bp was visible on agarose gels. PCR products of the ITS2 region of *D. leontinae* and *D. rapi* had lengths between 840 and 850 bp, respectively. Species-specific diagnostic patterns were generated with 11 different restriction enzymes but one enzyme (*Cfo*I) distinguished all the tested species of *Diadegma* (Wagener et al. 2004b). The maximum interspecific nucleotide divergence between *D. mollipla* and *D. semiclausum* was 6.8%, whereas the new *Diadegma* species exhibited about 35% divergence. The intraspecific divergence was 0.3 and 0.7% in *D. mollipla* and *D. semiclausum*, respectively (Wagener et al. 2004a).

Parasitoid strains seem to be associated with specific *P. xylostella* strains suggesting that effective biocontrol agents should be obtained from the area of origin of a pest strain. For example, two Ethiopian species of *Diadegma* did not parasitize French populations of DBM (Kirk et al. 2004). The Pakistani (P) and Romanian (R) *O. sokolowskii* were incompatible, and R × P could not produce any progeny indicating that they are two different strains of the same species (Kirk et al. 2004) and this inter-strain incompatibility may be due to certain symbionts such as *Wolbachia* (Rincon et al. 2004). If the release of an exotic biocontrol agent is necessary and compatible with the target host strain, then it is also important to evaluate its establishment and spread in the new habitat after its release.

Bottom-up effects on parasitoids

DBM parasitism by *D. insulare* was affected by the quality of both herbivore and its host plant. For example, ‘high quality’ DBM larvae feeding on plants grown at high nitrogen levels yielded 93% female wasps while ‘low quality’ DBM developing on plants grown at low nitrogen levels yielded 58% females (Fox et al. 1990). Different host plants can also have a significant impact on parasitism. Parasitism by *D. semiclausum* was greater with host larvae feeding on common cabbage (*Brassica*

oleracea (L.) var. *capitata*) than with larvae on cauliflower (*B. oleracea* var. *botrytis*), broccoli (*B. oleracea* var. *italica*) or Chinese cabbage (*B. campestris* L. ssp. *pekinensis*), but parasitism by *C. plutellae* was greatest with larvae present on Chinese cabbage compared with the other three host plants (Talekar & Yang 1991). These researchers stated that they could not provide any plausible explanation for such differences in parasitism. However, subsequent research revealed that plant volatiles from Chinese cabbage were more attractive to *C. plutellae* than those of common cabbage resulting in 4–15-fold higher parasitism of DBM larvae feeding on Chinese cabbage (Liu & Jiang 2003). In a choice test, DBM larvae feeding on sugar snap peas, *Pisum sativum* L. (Fabaceae) and collards, *B. oleracea* var. *acephala* were exposed to *D. mollipla*. Parasitism rate of DBM larvae on sugar snap peas was four times higher than that on collards and authors speculated that crucifer volatiles were unlikely to be used by this parasitoid for host location (Löhr & Rossbach 2004). Prior to this study, *D. mollipla* was recorded as an important parasitoid of potato tuber moth, *Phthorimaaea operculella* (Zeller) (Lepidoptera: Gelechiidae), on potato and tobacco in South Africa (Broodryk 1971) suggesting that *D. mollipla* is only loosely associated with the crucifers and DBM (Löhr & Rossbach 2004). Studies on other *Diadegma* species could generate more reliable results regarding any impact of such host-plant shift by *P. xylostella*.

Idris and Grafius (1996) assessed the impacts of various cultivated and wild crucifers on parasitism success of *D. insulare*. Highest DBM parasitism was recorded on *Sinapis arvensis* L. (= *Brassica kaber*) (91.5%) followed by *B. oleracea* var. *italica* (90.8%), *B. oleracea* var. *botrytis* (89.5%), *B. napus* L. (87.3%), *B. oleracea* var. *acephala* (83.3%), *B. nigra* (L.) (81.3%), *B. oleracea* var. *capitata* (79.5%), *Raphanus raphanistrum* L. (72.8%) *Erysimum cheiranthoides* L. (65.5%), *Thlaspi arvense* L. (65.0%), *Berteroa incana* (L.) (55.0%) and *Lepidium campestre* (L.) (39.3%). No parasitism occurred on *Capsella bursa-pastoris* (L.) and no DBM larva survived on *Barbarea vulgaris* R. Br. Developmental time of parasitized third instars to pupation was about 13 days on most of the tested plants. Larval developmental time increased on *E. cheiranthoides* (15.8 days), *B. incana* (16.0 days), *T. arvense* (16.3 days) and *L. campestre* (20.3 days). However, these results should be interpreted with caution as the nutrient compositions of the leaves used in this test might be quite different from each other given they were collected from different locations on the Michigan State University campus.

DBM natural enemies: Entomopathogens

Parasitoids are fundamental for an integrated and sustainable management program of *P. xylostella*, but supplemental control through the application of an insecticide is necessary when an economic threshold is exceeded (Shelton et al. 1983). Because of their specificity against target pests and minimal environmental impacts, microbial pest control agents (MPCAs) are welcome additions to IPM programs. The use of microbial insecticides in crop protection is increasing with the discovery and development of new entomopathogens. The potential of various entomopathogens including bacteria, fungi, viruses, protozoa, and nematodes as biocontrol agents has been tested for DBM management but few studies have demonstrated practical promising significance (Figure 1).

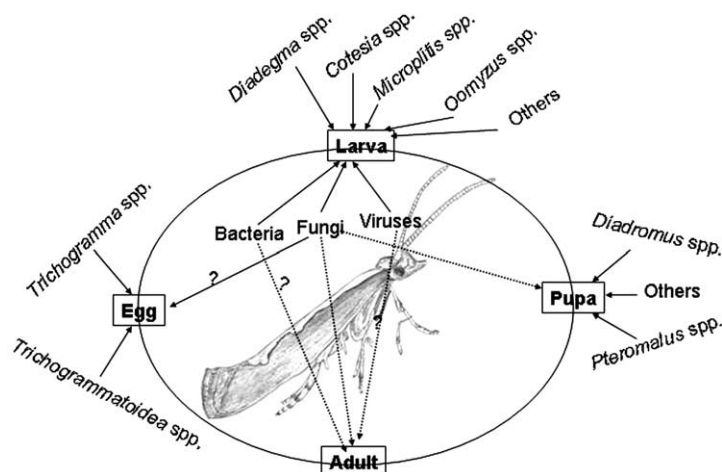


Figure 1. Principal biocontrol agents of *Plutell xylostella* (L.).

Bacteria

Bacillus thuringiensis Berliner (Bacillales: Bacillaceae). The Gram-positive soil bacterium, *Bacillus thuringiensis* (*Bt*), is a complex of subspecies characterized by production of parasporal crystalline inclusions during sporulation. These inclusions are comprised of relatively high quantities of one or more glycoproteins known as δ -endotoxins or insecticidal crystal proteins (ICPs) that play a vital role in *Bt* pathogenicity to insects and other invertebrates (Bauer 1995). Since the launch of the first commercial *Bt* product in France in 1938, over 100 *Bt*-based pesticides have been introduced across the world and currently δ -endotoxins are active ingredients in more than 90% of all MPCAs (Glare & O'Callaghan 2000). *Bt*-based products are the most promising alternatives to conventional insecticides because they are highly toxic to certain pests and are compatible with IPM strategies due to their narrow host specificity, high amenability to genetic engineering, and because they cause little or no harm to humans, most beneficial insects and other non-target organisms (Tabashnik 1994; Bauer 1995). In certain parts of the world, DBM has exhibited considerable resistance to *Bt*-based products (Sarfraz 2004). In principle, any interference with the cascade of steps associated with the mode of action of *Bt* Cry proteins (solubilization, proteolytic activation, passage through the peritrophic membrane, receptor binding, membrane insertion, pore formation, and osmotic lysis of midgut cells) may help the insect to develop resistance (Tabashnik 1994). Evidence exists that binding site alteration is a major mechanism involved in DBM resistance to *Bt* (Ferré & van Rie 2002; Sarfraz 2004).

The first case of field resistance to *Bt* was reported from Hawaii in 1990 when DBM field-collected populations displayed 30-fold resistance to Dipel® (*Btk*, *B. thuringiensis* subsp. *kurstaki*) (Tabashnik et al. 1990). Laboratory screening of this colony with Dipel® increased resistance rapidly and the resulting colony (NO-QA) was >6,800-fold resistant to Cry1Ac (Tabashnik et al. 1993). Tang et al. (1996) demonstrated that the Loxa-A colony from Florida was >1,500-fold resistant to Javelin® (*Btk* NRD12) in the second generation. In the absence of selection, resistance rapidly declined to about 300-fold but remained stable at this level in

subsequent generations. The insects with this stabilized level were more than 200-fold resistant to Cry1Aa, Cry1Ab, and Cry1Ac, but they were still fully susceptible to Cry1B and Cry1C. Resistance in DBM populations to *Btk* products resulted in heavy use of *Bta* (*B. thuringiensis* subsp. *aizawai*) that typically contains Cry1C, in addition to Cry1A toxins. Two Hawaiian colonies (NO-93, NO-95) displayed up to 20-fold resistance to Cry1Ca and were found to be only 2–4-fold resistant to *Bta* and 50–130-fold less susceptible to *Btk* formulations when compared with a susceptible colony (Liu et al. 2000). A field-collected population (Cry1C-Sel) from South Carolina displayed 31-fold resistance to Cry1C. Continuous laboratory selection, using first Cry1C protoxin, and in later generations transgenic broccoli expressing high levels of Cry1C, increased resistance to 12,400-fold (in neonates) and 63,100-fold (in second instars) after 26 generations (Zhao et al. 2000). A Malaysian colony (SERD3) was reported to be resistant to both *Btk* and *Bta*. After rearing in the laboratory for seven generations in the absence of any selection, DBM showed 330- and 160-fold resistance to *Btk* and *Bta*, respectively. Furthermore, selection during the next three generations with *Bta* increased resistance to *Bta* but only marginally to *Btk* and vice versa (Wright et al. 1997; Ferré & van Rie 2002).

Bt products are still effective against DBM in many crucifer producing regions and need to be used judiciously to conserve their efficacy. Braun et al. (2004) collected DBM from different sites in Saskatchewan, Canada in 1998 and 2001 and tested over 4600 larvae using commercial preparations of Dipel®. All the assayed larvae showed complete susceptibility to recommended field application rates.

Fungi

Bt and fungi are quite dissimilar in their mode of action. The ICPs and/or spores of the bacterium must be ingested by a larva to elicit an effect, while conidia of the fungus must contact insect cuticle, germinate, and penetrate it through enzymatic action and mechanical pressure, and initiate mycelial growth, resulting in death of insect in days (Glare & O'Callaghan 2000; Inglis et al. 2001). Several species of fungal pathogens, including *Zoophthora radicans* (Brefeld) Batko, *Beauveria bassiana* (Balsamo) Vuillemin, *Metarhizium anisopliae* (Metsch.) Sorokin, *Paecilomyces farinosus* (Holm ex Gray) Brown & Smith, *Fusarium* spp., *Pandora* spp., *Erynia* spp., *Conidiobolus* spp., and *Scopulariopsis* spp., have been isolated from DBM (Vandenberg et al. 1998; Cherry et al. 2004a; Kirk et al. 2004), but few have been studied in detail. Vandenberg et al. (1998) reported screening 55 isolates of different fungi including *B. bassiana*, *M. anisopliae*, *P. farinosus* and *Fusarium* sp. for DBM.

Zoophthora radicans (Brefeld) Batko (Zygomycetes: Entomophthorales). *Zoophthora* (= *Erynia*) *radicans*, an important entomopathogenic fungus isolated from DBM, is reported to cause epizootics under favorable environmental conditions and can reduce local populations to zero (Ooi 1981; Riethmacher et al. 1992). The moths infected/contaminated with *Z. radicans* can serve as a source of fungal inoculum in the field and airborne conidia lead to epizootics as a result of auto-dissemination in DBM populations (Vickers et al. 2004). Although infected insects can survive for some time, a reduction in feeding damage may occur sooner. For example, third instar infected larvae consumed 44% less foliage than healthy larvae (Furlong et al. 1997). Moths infected with this fungus laid significantly fewer eggs (Furlong et al. 1997) and

Z. radicans was also found to disrupt the mating behavior by inhibiting the response to and production of sex pheromone (Reddy et al. 1998).

Beauveria bassiana (Balsamo) Vuillemin (Hyphomycetes: Moniliaceae). There is increasing interest in the use of mycoinsecticides based on *Beauveria bassiana* for the control of DBM. This pathogen applied at the rate of 3×10^6 conidia mL⁻¹ gives 100% DBM mortality after 3–7 days (Furlong 2004), and contaminated moths effectively transmit (horizontal/passive transmission) this fungus to healthy moths and larvae foraging on plants. Sporulating cadavers producing *Z. radicans* or *B. bassiana* conidia result in similar transmission rates (Furlong and Pell 2001). Various commercial formulations of this fungus significantly suppressed the DBM populations in screened enclosures and the field (Shelton et al. 1998; Vandenberg et al. 1998; Becker 1999).

Other fungi. Jun (2000) tested infectivity of five isolates of *M. anisopliae* and one isolate of *Nomuraea rileyi* (Farlow) Sampson for *P. xylostella*. An isolate of *M. anisopliae* was highly infective with an LC₅₀ of 2.03×10^4 conidia mL⁻¹ and an LT₅₀ of 4.97 days at 10^7 conidia mL⁻¹. The *N. rileyi* gave high mortality but did not sporulate in any of the cadavers (Jun 2000). *Metarhizium anisopliae* isolated from DBM collected in eastern Romania was also very virulent (Kirk et al. 2004).

Viruses

A number of baculoviruses have been reported to infect *P. xylostella*. In a review of the potential viruses, Wilding (1986) concluded that granuloviruses (GVs) showed promising levels of pathogenicity. A Kenyan isolate of PxGV (Nya-01) applied in the field at 3.0×10^{13} occlusion bodies (OB) ha⁻¹ controlled DBM on kale with 82 and 90% infection rates for second and first instars, respectively (Grzywacz et al. 2004). Second instars of the Benin DBM strain were >90-fold more susceptible to this Nya-01 isolate than the Kenyan strain, and seedlings emerging from PxGV-loaded soils acquired adequate virus to infect DBM larvae feeding on them (Cherry et al. 2004b). The PxGv-fed larvae consumed less foliage and the proportion of decreasing total leaf consumption from the first through fourth instars was 93, 77, 53, and 46%, respectively, compared with their healthy counterparts (Lü et al. 2004). The PxGVs have been reported from various countries including Japan (Asayama & Osaki 1970), Taiwan (Kadir 1986), India (Rabindra et al. 1997), China (Kadir et al. 1999a) and Kenya (Grzywacz et al. 2004). Five Kenyan isolates of PxGV were found to be a mixture of different genotypes on the basis of sub-molar bands in their REN (restriction endonuclease) fragment profiles and their LD₅₀ values for DBM neonates were not significantly different from the Chinese, Japanese and Taiwanese strains (Woodward et al. 2004).

In China, a nucleopolyhedrovirus isolated from *P. xylostella* (PxMNPV) was characterized as genetically distinct from AcMNPV and AfMNPV. Based on relative differences in LC₅₀ values, PxMNPV was three to four log cycles more potent against DBM than either AcMNPV or AfMNPV (Kariuki & McIntosh 1999). The NPVs isolated from *Anagrapha falcifera* (Kirby) (AfMNPV), *Autographa californica* (Speyer) (AcMNPV), and *Galleria melonella* (L.) (GmMNPV) were also infectious to DBM

(Kadir et al. 1999b) but their potency was moderate to low, even after serial passage in this insect (Farrar & Ridway 1999).

A cypovirus (PxCPV) was isolated from a single *P. xylostella* larva in 1999 in an experimental field in Greater Accra region of Ghana. Like other CPVs, its replication was restricted to the midgut but it had very low virulence with an estimated LC_{95} value of $>9.0 \times 10^{10}$ OB mL^{-1} for third instar DBM larvae (Cherry et al. 2002b).

Other entomopathogens

Nematodes and microsporidia have also been reported to cause infection to DBM (Baur et al. 1997, 1998; Haque et al. 1999; Idris et al. 2002, 2004) but they need more extensive laboratory and field evaluations aimed at optimizing effectiveness. A nematode, *Steinernema carpocapsae* (Weiser), tested against DBM on *Nasturium aquaticum* (L.) Hayek farms in Hawaii gave 41% control of the pest alone and 58% when it was applied in conjunction with *Btk* (Baur et al. 1998).

A microsporidian, *Vairimorpha* sp., isolated from *P. xylostella* collected from Sundai Palas, Cameron Highlands, Malaysia caused 100% mortality at 1.5×10^3 spores per larva. The time to achieve 90–100% mortality was dose-dependent and varied from 5 days with 1.5×10^6 spores per larva to 11 days at 1.5×10^3 spores per larva (Haque et al. 1999). *Plutella xylostella* larval infection by a protozoan, *Nosema bombycis* Negali, was significantly higher in the Cameron Highlands (71.3%) than in the Serdang-Gombak lowlands (10%). Mortality was higher in younger instars (I and II) than older instars even at lower concentration of 4,260 spores μL^{-1} (Idris et al. 2004). Parasitoids such as *D. semiclausum* were involved in horizontal transmission of protozoan spores among DBM larvae (Idris et al. 2004).

Interactions between parasitoids and entomopathogens

When an insect pest is exposed to more than one killing agent, there is the possibility of an interaction that can enhance, limit, or limit and enhance various aspects of the effectiveness of control measures (Furlong & Pell 1996, 2000). Further, as the parasitoids actively forage for host larvae on crops, they can come into close contact with pathogens applied for pest control. It is, therefore, also important to investigate interactions which may occur between parasitoids and entomopathogens.

Bt toxins did not have detrimental impacts on adult *C. plutellae* (Chilcutt & Tabashnik 1999) and the braconid successfully completed its larval development in *Bt*-resistant DBM larvae feeding on *Bt* transgenic crucifers. *Bt*-resistant (BTR) *P. xylostella* larvae caused significantly greater damage to *Bt* leaves than *Bt*-susceptible (BTS) larvae and BTR-damaged leaves were more attractive to adult *C. plutellae* females than those damaged by BTS larvae. Of the 40 parasitoids tested, 27 landed on BTR-damaged leaves and only seven chose BTS-damaged leaves; six wasps did not respond. Parasitoid females did not distinguish between *Bt* and non-*Bt* plants both damaged by BTR larvae. For example, BTR-damaged *Bt* and non-*Bt* leaves were offered to 40 wasps; 19 flew to *Bt* leaves and 17 to non-*Bt* leaves, while four did not show any response. These findings suggest that *C. plutellae* is equally effective in controlling *Bt*-resistant DBM on genetically modified *Bt* plants and non-*Bt* plants (Schuler et al. 2003). *Bt*-derived commercial insecticides (Xentari® and Crymax®) were found safe for *D. insulare* adults and pupae (Xu et al. 2004).

Furlong and Pell (1996) observed that *Z. radicans* did not infect *C. plutellae*. However, its infection of DBM larvae ≤ 4 days after parasitism by *D. semiclausum* killed both host and the parasitoid (Furlong & Pell 2000). *Diadegma semiclausum* showed 100 times less susceptibility to *Z. radicans* than the DBM and has never been found infected in the field during epizootics (Pell et al. 2001). Increased concentrations ($> 1 \times 10^6$ conidia mL^{-1}) of *B. bassiana* showed significant adverse effects on *D. semiclausum* cocoon production and adult wasp emergence; parasitoid larvae within the host also became infected by the fungus (Furlong 2004). Microsporidia may affect parasitoids in different ways. For example, progeny of *Trichogramma chilonis* Ishii severely declined when they parasitized eggs of DBM infected with *Vairimorpha* species. The longevity and reproductive performance of parasitoids that emerged from infected hosts also reduced significantly (Schuld et al. 1999). *Nosema bombycis* infection altered diurnal activities of adult *D. semiclausum* and infected females spent significantly more time on activities that were not related to parasitism, for example, more moving (65%) than resting (25%), flying (5%), feeding (4%) and grooming (1%) (Idris et al. 2004).

Integration of biocontrol agents in IPM

Van Emden (1991) argued that insecticide concentrations could be reduced 3-fold on partially resistant host plants without appreciable increases in the pest population. Reducing pesticide usage in an insect pest management program can also be achieved through integration with biological control agents. Biocontrol of DBM is an important measure that can be integrated into modern IPM programs, particularly when the pest complex is simple and DBM is the dominant species (Hamilton et al. 2004).

Crucifer growers should choose cultivars that are partially resistant to DBM and more attractive to its natural enemies. They should avoid unnecessary use of fertilizers as it results in higher infestations of DBM and other crucifer pests such as *Delia* species (Dosdall et al. 2004a; Sarfraz et al., unpublished data). Based on pest scouting and monitoring, a selective insecticide (e.g., bacterial and/or fungal products, etc.) can be used to keep the pest population to a single action threshold (SAT) level (i.e., minimum acceptable pest infestation level) available to parasitoids. *Bt*-transgenic crops grown in conjunction with refugia may also enhance sustainable pest management. Refuge plants can serve as reservoirs for both the pest and its natural enemies. In theory, the majority of insects infesting *Bt* plants will be killed by *Bt* toxins and the survivors (*Bt*-resistant) may serve as hosts for parasitoids that will be present in refugia.

Mycoinsecticides are useful tools that are compatible with IPM of DBM depending on their target specificity. Their auto-dissemination with 'myco-sex pheromone traps' is currently being investigated elsewhere. It involves the attraction of male moths in response to synthetic female sex pheromone into specially designed inoculation chambers containing infective fungal spores. Once inside the chamber they become contaminated with conidia and then return to the crop, disseminating the pathogen in the local population (Vickers et al. 2004). Intensive research is needed to determine the potential and efficacy of other entomopathogens.

The design of an integrated approach requires that the characteristics of each control agent be documented carefully. Only with this knowledge can the most effective strategy be employed. For example, *C. plutellae* is less efficient at host

searching, prone to multiparasitism, and DBM larvae parasitized by this braconid live for longer periods resulting in more foliage consumption than in unparasitized counterparts. Current data indicate an approach that, whenever possible, larval parasitoids, in particular a *Diadegma* species, and pupal parasitoids such as *Diadromus* species, should be introduced, keeping in mind the native parasitoid complex in a given region. Application of *Bt* toxins or other selective insecticides should be made to suppress the pest population below the SAT-level. Remaining individuals will provide hosts for the larval parasitoids, and successfully pupating insects will serve as hosts for pupal parasitoids.

Growers need both the information and tools to manage DBM and other pests. The co-ordinated use of biocontrol agents can lead to effective pest control. Overall, this strategy will help maintain the viability of crucifers in many regions especially where resistance to insecticides is common.

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